MODULE 2: ENERGY DEVICES & SEMICONDUCTOR ELECTRODES

Complete Unabridged Course Notes, Derivations, Diagrams & Topic-Wise CAT/FAT Solved Problems

Fully Aligned with Official VIT Faculty Lecture Decks (72 Slides), Department Question Bank, and Past FAT Papers (CO: 2)

Module 2 Detailed Syllabus Breakdown (100% Complete Coverage):

Section 1: Electrochemical & Electrolytic Cells: Galvanic vs Electrolytic cells, LOAN rule, Daniell Cell, EMF, Salt Bridge construction & 3 functions, Nernst Equation derivation, Equilibrium constant (K_{eq}).
Section 2: Industrial Electrolysis: Downs cell for molten $NaCl$ extraction (chemistry, $CaCl_2$ flux, wire gauze diaphragm), Water splitting electrolysis (acidic vs alkaline mechanisms, OER/HER).
Section 3: Semiconductor Electrode Materials: Band theory, effect of temperature on conductivity, Intrinsic vs Extrinsic (n-type & p-type), Monocrystalline vs Polycrystalline vs Amorphous Silicon.
Section 4: Silicon Purification (9N Semiconductor Grade): Siemens Process (distillation of $SiHCl_3$), Zone Refining (segregation coefficient $k < 1$), Czochralski (CZ) single-crystal pulling technique.
Section 5: Advanced Batteries: Primary vs Secondary vs Fuel Cells, Lithium-ion battery (LIB) chemistry, why lithium, rocking-chair mechanism, charging/discharging reactions, variants (LFP, NMC, Li-poly), BMS safety & thermal runaway.
Section 6: Fuel Cells: Working principles, Proton Exchange Membrane Fuel Cell (PEMFC, Nafion, Pt/C catalyst), Solid Oxide Fuel Cell (SOFC, YSZ electrolyte, Ni cermet anode, LSM cathode, fuel flexibility).
Section 7: Solar Energy Conversion: Photovoltaic (PV) cells, Photoelectrochemical (PEC) cells, Dye-Sensitized Solar Cells (DSSC / Grätzel Cell) sandwich architecture, 4-step mechanism, Ru-dyes, electrolyte regeneration, PV vs PEC vs DSSC comparison.

1. Electrochemical and Electrolytic Cells

Electrochemical devices convert chemical energy directly into electrical energy or utilize electrical energy to drive non-spontaneous chemical transformations.

1.1 Galvanic (Electrochemical) vs. Electrolytic Cells

Parameter	Galvanic / Voltaic Cell	Electrolytic Cell
Energy Conversion	Converts spontaneous chemical energy into electrical energy ($\Delta G < 0, E_{cell} > 0$).	Consumes external electrical energy to drive non-spontaneous reactions ($\Delta G > 0, E_{ext} > E^{\circ}_{cell}$).
Anode Polarity & Reaction	**Negative (−).** Oxidation occurs (LOAN: Left, Oxidation, Anode, Negative).	**Positive (+).** Oxidation occurs (connected to positive terminal of DC power source).
Cathode Polarity & Reaction	**Positive (+).** Reduction occurs (electrons flow into cathode from external circuit).	**Negative (−).** Reduction occurs (connected to negative terminal of DC power source).
Electron Flow Direction	Electrons flow spontaneously from anode to cathode in external circuit.	Electrons pushed from anode to cathode by external power supply.
Salt Bridge Requirement	Requires a salt bridge or porous partition to maintain electrical contact.	Usually uses a single electrolyte compartment; no salt bridge required.
Physical Examples	Daniell cell, Dry cell, Lead-acid battery (during discharge), Li-ion battery.	Downs cell, Hall-Héroult cell, Electroplating baths, Water electrolysis.

1.2 The Daniell Cell: Construction and Architecture

Invented by John Frederic Daniell in 1836, the cell consists of:

- **Anode Half-Cell:** Metallic Zinc (Zn) rod immersed in 1.0 M $ZnSO_4$ aqueous solution.
- **Cathode Half-Cell:** Metallic Copper (Cu) rod immersed in 1.0 M $CuSO_4$ aqueous solution.
- **Salt Bridge:** Inverted U-tube containing agar-agar gel saturated with inert KCl or KNO_3.

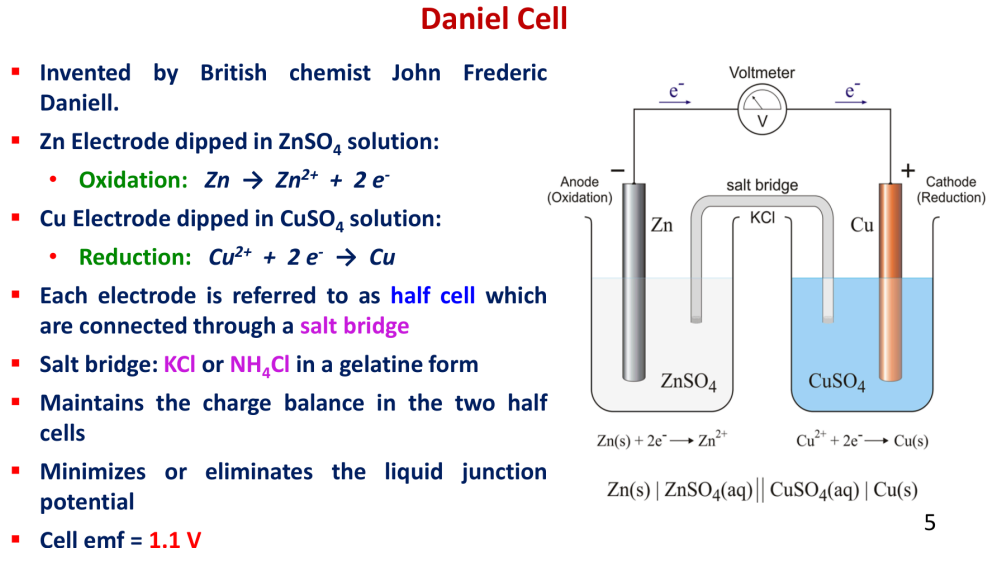


Figure 2.1: Construction, electrode half-cells, salt bridge, and redox electron flow in the standard Daniell Galvanic Cell.

- **Overall Cell Reaction:** $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$
- **Standard Cell Potential:**

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}} = E^{\circ}_{C\frac{u^{2+}}{C}u} - E^{\circ}_{Z\frac{n^{2+}}{Z}n} = +0.34\ \text{V} - (-0.76\ \text{V}) = \mathbf{+1.10\ V}$$

- **IUPAC Cell Representation:** $Zn(s) \mid Zn^{2+}(1\ \text{M}) \parallel Cu^{2+}(1\ \text{M}) \mid Cu(s)$

1.3 The Salt Bridge: Construction and 3 Critical Functions

A glass U-tube filled with a warm semi-solid gel of agar-agar or gelatin boiled with a concentrated solution of an inert electrolyte (KCl, KNO_3, or NH_4NO_3).

- **Criterion for Electrolyte Selection:** The ionic velocities (mobilities) of the cation and anion must be virtually identical ($u_{K^+} \approx u_{Cl^-}$).
- **Three Essential Functions:**
  1. **Completes the Internal Electric Circuit:** Permits migration of ions between the two half-cells, allowing current to flow continuously through the external circuit.
  2. **Maintains Electrical Neutrality:** Prevents charge accumulation in half-cells. As Zn^{2+} accumulates at the anode, Cl^- anions migrate in to neutralize positive charge. As Cu^{2+} is depleted at the cathode, K^+ cations migrate in to neutralize excess SO_4^{2-} ions.
  3. **Eliminates Liquid Junction Potential ($E_j \approx 0$):** Because K^+ and Cl^- diffuse at nearly identical speeds, no differential charge buildup occurs at the liquid-liquid boundary, keeping junction potential negligible.

1.4 Thermodynamic Derivation of the Nernst Equation

From chemical thermodynamics, the non-expansion useful work performed by an electrochemical cell is equal to the decrease in Gibbs free energy:

$$\Delta G = -nFE_{\text{cell}} \quad \text{and} \quad \Delta G^{\circ} = -nFE^{\circ}_{\text{cell}}$$

From the van 't Hoff reaction isotherm:

$$\Delta G = \Delta G^{\circ} + RT \ln Q$$

where Q is the reaction quotient. Substituting $-nFE$:

$$-nFE_{\text{cell}} = -nFE^{\circ}_{\text{cell}} + RT\ln Q$$

Dividing both sides by $-nF$:

$$\boldsymbol{E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{RT}{nF}\ln Q}$$

Converting to common base-10 logarithm ($\ln Q = 2.303\log_{10} Q$):

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{2.303RT}{nF}\log_{10} Q$$

At standard laboratory temperature ($T = 298.15$ K, $R = 8.314$J K^{-1}mol^{-1}, $F = 96485$C mol^{-1}):

$$\frac{2.303 \times 8.314 \times 298.15}{96485} = 0.05916 \text{ V}$$

$$\boldsymbol{E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.05916}{n}\log_{10} Q}$$

- **Equilibrium Constant (K_{eq}):** At dynamic equilibrium, cell EMF becomes zero ($E_{\text{cell}} = 0$) and $Q = K_{\text{eq}}$:

$$0 = E^{\circ}_{\text{cell}} - \frac{0.05916}{n}\log_{10} K_{\text{eq}} \Longrightarrow \boldsymbol{\log_{10} K_{\text{eq}} = \frac{nE^{\circ}_{\text{cell}}}{0.05916}}$$

[CAT-1 / FAT Exam Problem — 5 Marks]
Problem: Calculate the EMF of the following cell at 298 K:
$Cd(s) \mid Cd^{2+}(0.01\text{ M}) \parallel Ag^{+}(0.1\text{ M}) \mid Ag(s)$
Given: $E^{\circ}_{C\frac{d^{2+}}{C}d} = -0.40$ V, $E^{\circ}_{A\frac{g^{+}}{A}g} = +0.80$ V.
Step-by-Step Model Solution:
1. **Identify Half-Reactions and n:**
   - Anode (Oxidation): $Cd(s) \rightarrow Cd^{2+}(aq) + 2e^{-}$
   - Cathode (Reduction): $2Ag^{+}(aq) + 2e^{-} \rightarrow 2Ag(s)$
   - Overall Reaction: $Cd(s) + 2Ag^{+}(aq) \rightarrow Cd^{2+}(aq) + 2Ag(s) \Longrightarrow \boldsymbol{n = 2}$
2. **Standard EMF (E°_{cell}):**

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{cathode}} - E^{\circ}_{\text{anode}} = +0.80\text{ V} - (-0.40\text{ V}) = \boldsymbol{+1.20\text{ V}}$$

3. **Reaction Quotient (Q):**

$$Q = \frac{[Cd^{2+}]}{[Ag^{+}]^2} = \frac{0.01}{(0.1)^2} = \frac{0.01}{0.01} = \boldsymbol{1.0}$$

4. **Apply Nernst Equation:**

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.05916}{2}\log_{10}(1.0) = 1.20\text{ V} - 0 = \boldsymbol{1.20\text{ V}}$$

2. Industrial Electrolysis: Downs Cell & Water Splitting

2.1 The Downs Cell for Sodium Extraction

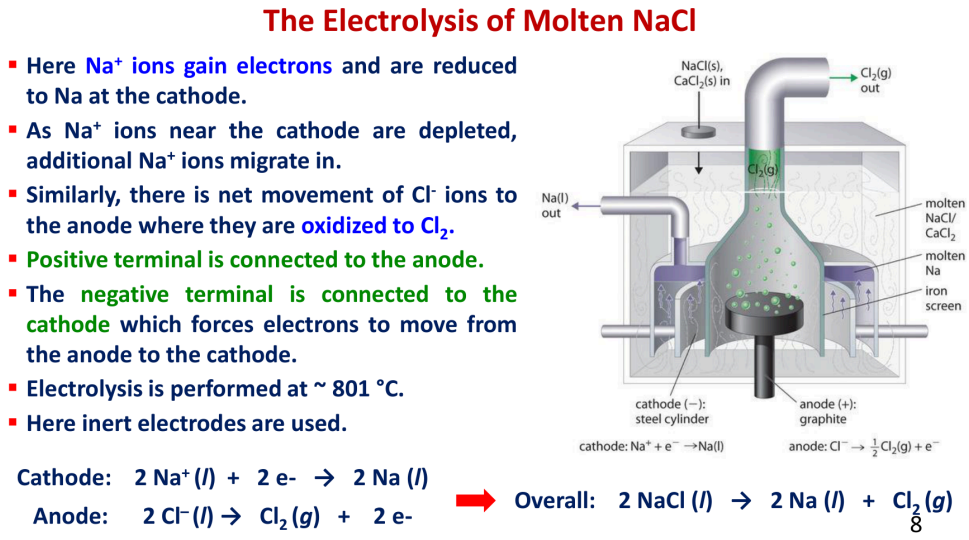


Figure 2.2: Industrial Downs Cell architecture showing central graphite anode, steel cylindrical cathode, and steel wire gauze diaphragm for molten NaCl electrolysis.

The commercial electrochemical extraction of metallic sodium by the electrolysis of molten sodium chloride ($NaCl$).

- **Raw Material Mixture:** Molten $NaCl$ (40%) fluxed with Calcium Chloride ($CaCl_2$, 60%).
  - **Function of $CaCl_2$ Flux:** Lowers the melting point of the electrolyte from 801° C (pure $NaCl$) down to $\approx 580 - 600$° C. This prevents sodium metal from vaporizing (bp of Na is 883° C) and prevents sodium from forming a colloidal fog in the melt.
- **Downs Cell Construction:**
  - **Anode:** Centrally positioned cylindrical high-density graphite rod.
  - **Cathode:** Concentric circular ring of steel surrounding the graphite anode.
  - **Diaphragm:** Cylindrical steel wire gauze screen placed between anode and cathode to prevent newly formed liquid sodium from contacting chlorine gas and recombining explosively.
  - **Collection Dome & Pipe:** An inverted iron hood captures dry chlorine gas (Cl_2), while molten sodium, being less dense than the molten salt mixture ($d_{\text{Na}} = 0.8\text{g/cm}^3 < d_{\text{melt}} = 2.1\text{g/cm}^3$), floats upward into an inverted annular collector and is piped into storage vessels under mineral oil.
- **Electrode Reactions:**
  - **Anode (Oxidation):** $2Cl^-(l) \rightarrow Cl_2(g) + 2e^-$
  - **Cathode (Reduction):** $2Na^+(l) + 2e^- \rightarrow 2Na(l)$
  - **Overall Cell Reaction:** $2NaCl(l) \rightarrow 2Na(l) + Cl_2(g)$
  - Operating cell voltage: $\approx 4.0 - 4.5$ V; Operating current: up to 40000 A.

2.2 Electrolytic Decomposition of Water (Water Splitting)

Electrolytic Decomposition of Water

- A pair of inert electrodes are dipped into the solution and applying a voltage between them results in the rapid evolution of bubbles of H_2 and O_2.
- Due to very poor electrical conductivity of pure water, a small amount of an ionic solute (such as H_2SO_4 or Na_2SO_4) is added to increase its electrical conductivity.
- The electrolytes are chosen such a way that the ions that are harder to oxidize or reduce than water:

Cathode: $4\,H^+(aq) + 4\,e^- \rightarrow 2\,H_2(g)$
Anode: $2\,H_2O \rightarrow O_2(g) + 4\,H^+ + 4\,e^-$
Overall: $2\,H_2O \rightarrow O_2(g) + 2\,H_2(g)$

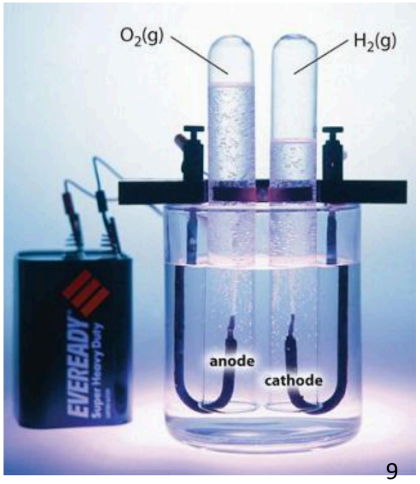


Figure 2.3: Water splitting electrolytic cell showing stoichiometric decomposition into hydrogen and oxygen (2 : 1 volume ratio).

The stoichiometric decomposition of water into hydrogen and oxygen gases:

$$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$$

- **Theoretical Decomposition Voltage:** $E^\circ_{\text{cell}} = 1.23$ V at 298 K (practical overvoltage requires $1.8 - 2.0$ V).
- **Acidic Electrolysis (Proton Exchange / H_2SO_4):**
  - Anode (OER — Oxygen Evolution Reaction): $2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$ ($E^\circ = +1.23$ V)
  - Cathode (HER — Hydrogen Evolution Reaction): $4H^+(aq) + 4e^- \rightarrow 2H_2(g)$ ($E^\circ = 0.00$ V)
- **Alkaline Electrolysis ($20 - 30\%\ KOH$, Nickel Mesh Electrodes):**
  - Anode: $4OH^-(aq) \rightarrow O_2(g) + 2H_2O(l) + 4e^-$ ($E^\circ = +0.40$ V)
  - Cathode: $4H_2O(l) + 4e^- \rightarrow 2H_2(g) + 4OH^-(aq)$ ($E^\circ = -0.83$ V)
  - Volume ratio produced: Exactly 2 volumes H_2 : 1 volume O_2.

3. Semiconductor Electrode Materials & Band Theory

A semiconductor is a solid whose electrical conductivity lies between that of a metallic conductor ($\sigma > 10^4$S cm^{-1}) and an electrical insulator ($\sigma < 10^{-10}$S cm^{-1}).

3.1 Energy Band Theory

Electrode materials: Semiconductors

- A semiconductor is a substance, usually a solid chemical element or compound that can conduct electricity under some conditions but not others, making it a good medium for the control of electrical current.
- It has almost filled valence band, empty conduction band and very narrow energy gap i.e., of the order of 1 eV. Energy gap of Silicon (Si) and Germanium (Ge) are 1.0 and 0.7 eV respectively. Consequently Si and Ge are semiconductors.

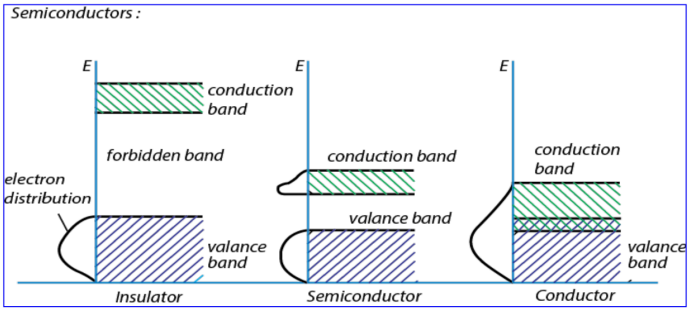


Figure 2.4: Energy band structure and temperature dependence of electrical conductivity in semiconductors.

- **Valence Band (VB):** The highest range of electron energies in which electrons are present in the ground state.
- **Conduction Band (CB):** The lowest range of vacant, unoccupied electron energy levels.
- **Forbidden Bandgap (E_g):** The forbidden energy separation between the top of the VB and bottom of the CB:
  - **Metals:** VB and CB overlap ($E_g \approx 0$ eV). Large electron mobility.
  - **Semiconductors:** Moderate bandgap (0.5 eV $< E_g <$ 3.0 eV, e.g., Silicon $E_g = 1.12$ eV, Germanium $E_g = 0.67$ eV).
  - **Insulators:** Extremely wide forbidden bandgap ($E_g > 5.0$ eV, e.g., Diamond $E_g = 5.47$ eV).

3.2 Temperature Dependence of Semiconductor Conductivity

- **At Absolute Zero ($T = 0$ K):** All electrons are locked in covalent bonds in the VB. CB is completely vacant. The semiconductor behaves as a **perfect electrical insulator** ($\sigma = 0$).
- **At Room Temperature ($T = 300$ K):** Thermal kinetic energy excites a fraction of electrons across E_g into the CB, leaving behind equal numbers of empty electron vacancies (**holes**) in the VB. Both free electrons (n_e) and positive holes (n_h) contribute to electrical conduction.
- **Conductivity Equation:** $\sigma = e(n_e\mu_e + n_h\mu_h)$, where μ is carrier mobility.
- **Temperature Coefficient:** Unlike metals (which have positive resistance temperature coefficient $\alpha > 0$), semiconductors exhibit a **negative temperature coefficient of resistance** ($\alpha < 0$); resistance decreases exponentially as temperature rises: $R(T) = R_0 \exp\left(\frac{E_g}{2}k_BT\right)$.

3.3 Intrinsic vs. Extrinsic Semiconductors

- **Intrinsic Semiconductors:** Chemically pure elements (Group 14: Si, Ge) with no dopant atoms ($n_e = n_h = n_i$).
- **Extrinsic Semiconductors:** Intrinsic semiconductors doped with trace donor or acceptor impurities (1 in 10^6 atoms):
  - **n-type Semiconductor:** Doped with pentavalent Group 15 donors (Phosphorus P, Arsenic As, Antimony Sb). Four valence electrons form covalent bonds with adjacent Si atoms; the 5th electron is loosely bound ($E_d \approx 0.05$ eV) and readily enters the CB. Majority carriers: **electrons** ($n_e \gg n_h$). Fermi level (E_F) shifts upward close to CB.
  - **p-type Semiconductor:** Doped with trivalent Group 13 acceptors (Boron B, Aluminum Al, Gallium Ga). Boron forms 3 covalent bonds, leaving an unfilled orbital (hole) in the lattice. An electron from an adjacent bond readily hops in, creating a mobile positive hole in the VB. Majority carriers: **holes** ($n_h \gg n_e$). Fermi level (E_F) shifts downward close to VB.

3.4 Comparison of Silicon Structural Forms

Silicon Type	Lattice Structure	Electronic Properties	PV Efficiency
Monocrystalline (c-Si)	Continuous single diamond crystal lattice; zero grain boundaries.	Highest electron mobility ($\mu_e \approx 1400 \frac{\text{cm}^2}{\text{V s}}$); low recombination.	20% – 26%
Polycrystalline (poly-Si)	Aggregate of multi-oriented crystallites separated by grain boundaries.	Grain boundaries act as charge recombination centers; lower carrier lifetime.	15% – 18%
Amorphous (a-Si:H)	Non-crystalline random network; dangling bonds passivated by hydrogen.	Direct bandgap (1.7 eV); high optical absorption; Staebler-Wronski light degradation.	8% – 12%

4. Three-Stage Silicon Purification (9N Electronic Grade)

High-efficiency semiconductor electronics require electronic-grade silicon (EGS) with purity $> 99.9999999\%$ (less than 1 impurity atom per billion silicon atoms).

4.1 Stage 1: Chemical Purification (Siemens Process)

1. **Carbothermic Reduction:** Quartzite silica is reduced in an electric arc furnace at 1900° C:

$$SiO_2(s) + 2C(s) \rightarrow Si(\text{metallurgical grade}, 98\%) + 2CO(g)$$

2. **Hydrochlorination:** Powdered MGS reacts with dry anhydrous HCl in a fluidized bed reactor at 300° C:

$$Si(s) + 3HCl(g) \rightarrow SiHCl_3(l, \text{trichlorosilane}) + H_2(g)$$

3. **Fractional Distillation:** Trichlorosilane ($SiHCl_3$) is a volatile liquid with boiling point 31.8° C. Multiple fractional distillation columns remove volatile impurities ($BCl_3, PCl_3, AsCl_3, FeCl_3$).
4. **Chemical Vapor Deposition (CVD):** High-purity $SiHCl_3$ vapor mixed with H_2 is decomposed onto pure heated silicon filament rods at 1100° C, yielding electronic-grade polycrystalline silicon:

$$SiHCl_3(g) + H_2(g) \rightarrow Si(s, \text{electronic grade}) + 3HCl(g)$$

4.2 Stage 2: Zone Refining Technique

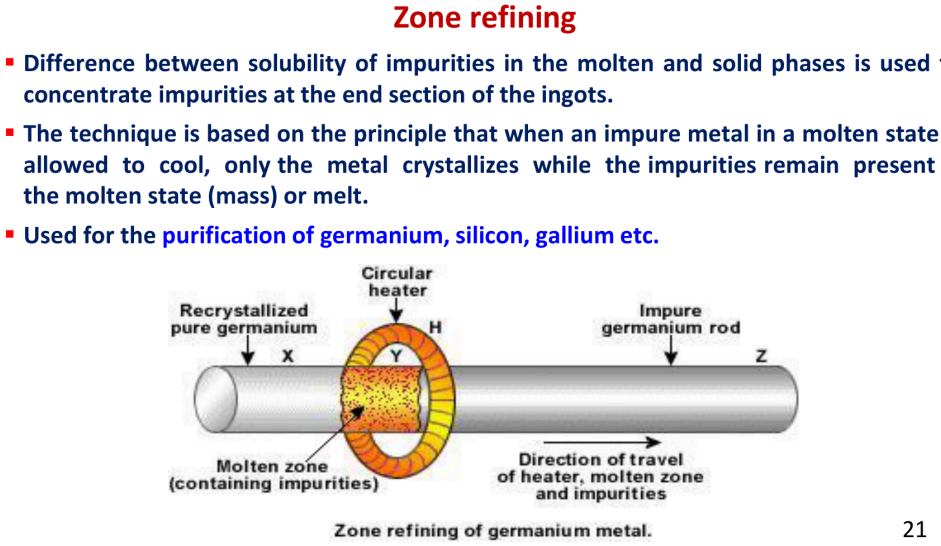


Figure 2.5: Zone refining apparatus showing moving RF induction heater coil sweeping impurities along molten zone.

Invented by William Gardner Pfann at Bell Labs, zone refining utilizes the physical principle that **impurities are significantly more soluble in the liquid (molten) phase than in the solid phase**.

- **Segregation Coefficient (k):** Defined as the ratio of impurity concentration in the freezing solid (C_s) to that in the molten liquid (C_l):

$$k = \frac{C_s}{C_l}$$

  For almost all impurities in silicon (B, P, Fe, Al), $k < 1$ (typically $k = 0.001 - 0.35$).
- **Refining Operation:**
  - A long cylindrical ingot of polycrystalline silicon is placed inside a quartz tube under an inert Argon atmosphere.
  - A circular high-frequency radio-frequency (RF) induction heating coil creates a narrow molten zone ($\approx 2 - 3$ cm wide).
  - The heating coil moves very slowly from one end of the ingot to the other.
  - As the molten zone traverses the rod, pure silicon crystallizes behind it ($C_s = kC_l < C_l$), while impurities selectively partition into the liquid melt.
  - The molten zone sweeps impurities along to the trailing end of the silicon bar.

- Repeating multiple passes (10 to 20 passes) in the same direction concentrates all impurities at the terminal end, which is cropped off and discarded, leaving silicon of 99.9999999% (9N) purity.

4.3 Stage 3: Czochralski (CZ) Single-Crystal Pulling

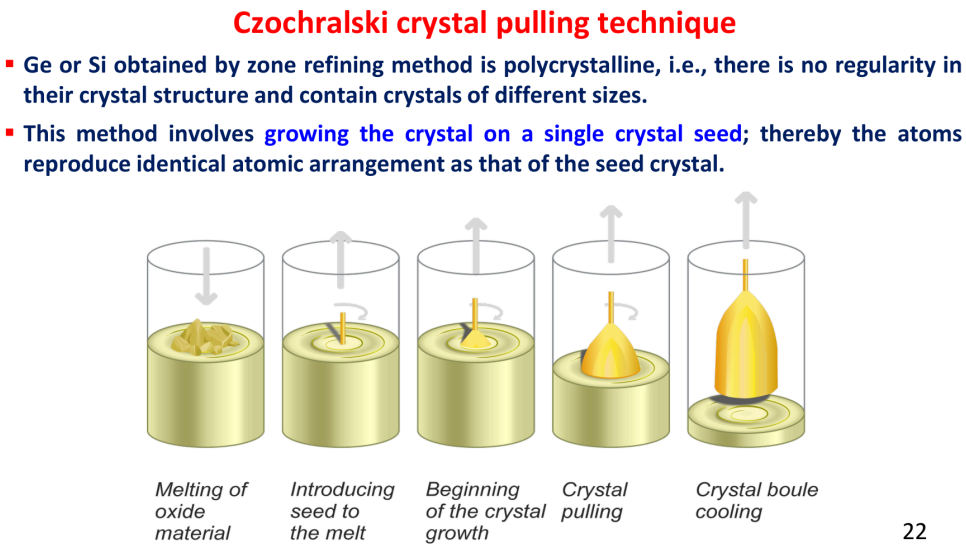


Figure 2.6: Czochralski crystal pulling furnace showing rotating seed crystal pulling dislocation-free silicon boule from melt at 1425° C.

Zone-refined silicon is polycrystalline. To make microchips and high-efficiency solar cells, it must be converted into a continuous single crystal.

- **Apparatus & Operation:**
  - Electronic-grade polysilicon is placed in a high-purity fused silica quartz crucible inside an argon-purged vacuum furnace and heated to 1425° C (silicon melting point 1414° C).
  - A small, perfectly oriented single-crystal seed (oriented along $< 100 >$ or $< 111 >$) is lowered on a pulling shaft and touched to the surface of the molten silicon.
  - The seed crystal is slowly pulled upward at a rate of $1 - 2$ mm/min while rotating clockwise; the crucible rotates counter-clockwise.
  - As the seed is pulled, molten silicon freezes onto it, replicating the exact crystalline lattice orientation of the seed.
  - The resulting massive, cylindrical single-crystal ingot (boule) measures up to 300 mm (12 inches) in diameter, over 2 meters in length, and weighs > 250 kg.

5. Advanced Secondary Batteries: The Lithium-Ion Battery (LIB)

5.1 Primary vs. Secondary vs. Fuel Cells

Property	Primary Battery	Secondary Battery	Fuel Cell
Reversibility	Irreversible electrochemical redox; discarded after single discharge.	Reversible redox; electrically recharged by passing DC current in opposite direction.	Continuous electrochemical conversion; never discharges as long as fuel is supplied.
Reactant Location	Stored internally within electrodes.	Stored internally within active electrode materials.	Supplied continuously from external storage tanks (H_2, O_2).
Life Cycle	Single use (1 cycle).	Long cycle life ($500 - 3000$ charge-discharge cycles).	Continuous operation indefinitely (limited only by catalyst/membrane degradation).

Examples	Zinc-carbon Leclanché cell, Alkaline $Zn-MnO_2$.	Lithium-ion $(Li-\text{ion})$, Lead-acid, $Ni-MH$.	PEMFC, SOFC, Alkaline Fuel Cell (AFC).

5.2 Why Lithium? Fundamental Advantages

1. **Lightest Metal:** Molar mass $M = 6.94\text{g mol}^{-1}$; density $\rho = 0.534\text{g cm}^{-3}$ (floats on water).
2. **Highest Standard Oxidation Potential:** Standard reduction potential $E^{\circ} = -3.04$ V (most electropositive metal), enabling single cell voltages of $3.6-3.8$ V (triple that of Ni-MH or Lead-acid).
3. **Highest Theoretical Specific Capacity:** 3860mAh g^{-1} (compared to Zn: 820mAh g^{-1}, Pb: 260mAh g^{-1}).
4. **Small Ionic Radius ($Li^+ = 0.76$ Å):** Enables rapid solid-state diffusion and reversible intercalation into host crystal lattices without severe structural degradation.

5.3 Construction and Architecture of a Lithium-Ion Cell

- **Cathode (Positive Electrode):** Layered transition metal oxide (typically Lithium Cobalt Oxide, $LiCoO_2$, or Lithium Iron Phosphate, $LiFePO_4$) coated onto an **Aluminium foil** current collector.
- **Anode (Negative Electrode):** Layered graphitic carbon (Li_xC_6) coated onto a **Copper foil** current collector.
- **Electrolyte:** 1.0 M Lithium Hexafluorophosphate ($LiPF_6$) salt dissolved in a non-aqueous, anhydrous organic carbonate mixture: Ethylene Carbonate (EC) + Dimethyl Carbonate (DMC). (Water cannot be used because lithium reacts vigorously with H_2O).
- **Separator:** Microporous polyethylene / polypropylene membrane ($\approx 15-25\mu$ m) allowing free passage of Li^+ ions while preventing physical electrical contact between anode and cathode.

5.4 The "Rocking-Chair" Intercalation Mechanism

Lithium-ion batteries operate via insertion and extraction (**intercalation/de-intercalation**) of Li^+ ions shuttling back and forth across the electrolyte:

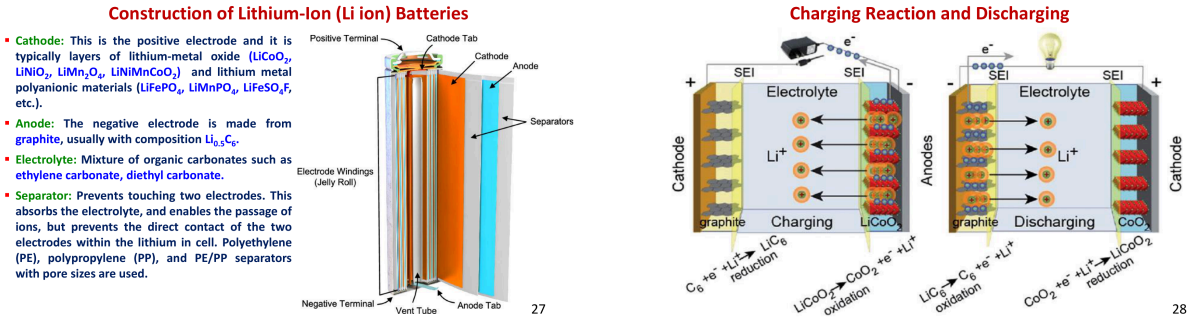


Figure 2.7: (Left) Lithium-ion cell internal construction and layered electrodes; (Right) Rocking-chair Li^+ intercalation/de-intercalation charge and discharge mechanisms.

- **Discharge Reactions (Spontaneous Cell Operation, $E_{\text{cell}} \approx 3.7$ V):**
  - **Anode (Oxidation):** $Li_xC_6 \rightarrow 6C + xLi^+ + xe^-$
  - **Cathode (Reduction):** $Li_{1-x}CoO_2 + xLi^+ + xe^- \rightarrow LiCoO_2$
  - **Net Cell Reaction:** $Li_xC_6 + Li_{1-x}CoO_2 \rightarrow 6C + LiCoO_2$
- **Charging Reactions (Driven by External DC Power Source):**
  - **Cathode (Oxidation):** $LiCoO_2 \rightarrow Li_{1-x}CoO_2 + xLi^+ + xe^-$
  - **Anode (Reduction):** $6C + xLi^+ + xe^- \rightarrow Li_xC_6$

5.5 Safety Hazards, Degradation, and BMS Protection

- **Thermal Runaway:** Overcharging, internal micro-shorts, or mechanical crushing causes local heating ($T > 130^{\circ}$ C). The Solid Electrolyte Interphase (SEI) decomposes exothermically, followed by cathode decomposition releasing oxygen gas (O_2), which reacts violently with flammable organic carbonate solvents, leading to smoke, fire, and venting explosion.

- **Lithium Dendrite Growth:** Fast charging at low temperatures causes metallic lithium plating onto the graphite anode surface. Needle-like dendritic crystals pierce the polymer separator, causing catastrophic internal short circuits.
- **Battery Management System (BMS):** Electronic protection circuit continuously monitoring individual cell voltages, temperature, and charge/discharge currents, executing emergency cutoff if voltage exceeds 4.25 V or temperature exceeds 60° C.

6. Fuel Cells: PEMFC and SOFC

A fuel cell is an electrochemical energy conversion device that converts the chemical energy of a fuel (H_2, natural gas) and an oxidant (O_2, air) directly into electrical energy via continuous catalytic redox reactions.

6.1 Proton Exchange Membrane Fuel Cell (PEMFC)

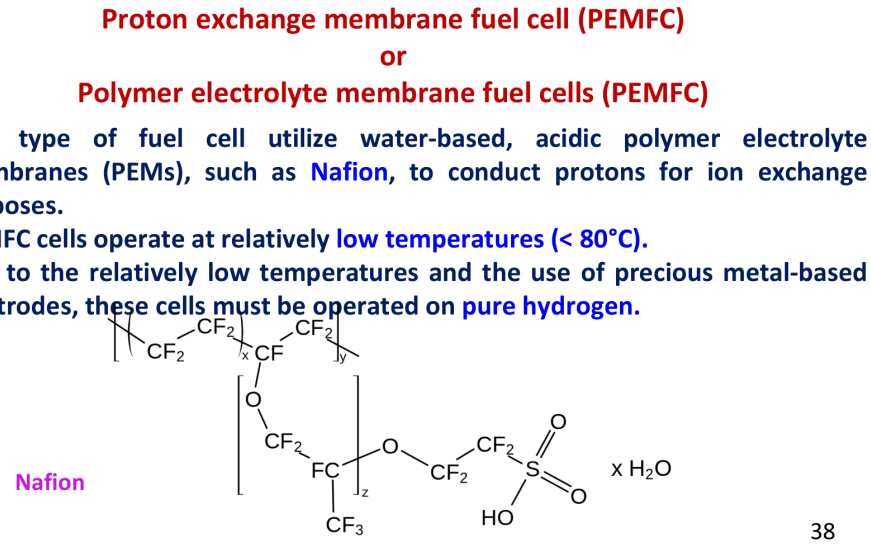


Figure 2.8: Proton Exchange Membrane Fuel Cell (PEMFC) construction showing Nafion membrane, Pt/C catalysts, and gas diffusion channels.

- **Operating Temperature:** $60 - 80°$ C (Low temperature; enables rapid startup within seconds).
- **Solid Polymer Electrolyte:** Perfluorosulfonic acid membrane (**Nafion**), which selectively conducts hydrated H^+ protons via sulfonic acid groups ($-SO_3^- H^+$) while blocking electrons and reactant gases.
- **Electrode Catalysts:** High-surface-area Platinum nanoparticles dispersed on carbon black ($P\frac{t}{C}$) hot-pressed onto porous carbon cloth/paper gas diffusion layers (GDL).
- **Electrode Reactions:**
  - **Anode (Oxidation):** $2H_2(g) \xrightarrow{\text{Pt}} 4H^+ + 4e^-$ ($E^\circ = 0.00$ V)
  - **Cathode (Reduction):** $O_2(g) + 4H^+ + 4e^- \xrightarrow{\text{Pt}} 2H_2O(l)$ ($E^\circ = +1.23$ V)
  - **Overall Cell Reaction:** $2H_2(g) + O_2(g) \rightarrow 2H_2O(l) + \text{Electricity} + \text{Heat}$
- **Advantages:** High power density ($\approx 1\text{W/cm}^2$), low operating temperature, zero tailpipe emissions (pure water byproduct), ideal for fuel cell electric vehicles (FCEVs).
- **Disadvantages:** Highly expensive platinum catalyst; severe poisoning of Pt catalyst by trace carbon monoxide ($CO > 10$ ppm); membrane requires delicate humidification management.

6.2 Solid Oxide Fuel Cell (SOFC)

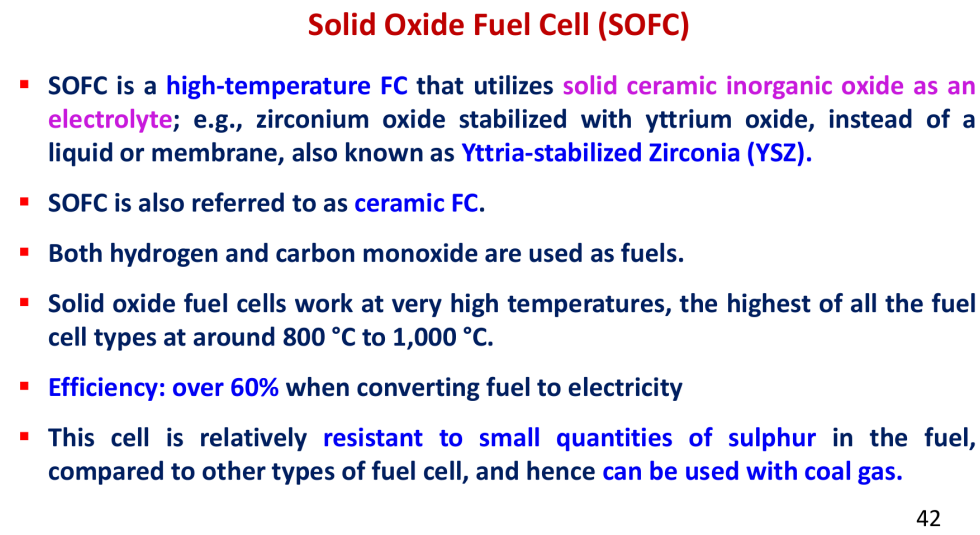


Figure 2.9: Solid Oxide Fuel Cell (SOFC) ceramic architecture showing YSZ solid electrolyte conducting $O^{\{2-\}}$ oxide ions at $800-1000^\circ$ C.

- **Operating Temperature:** $700-1000^\circ$ C (High temperature; solid ceramic materials).
- **Solid Ceramic Electrolyte:** Dense, non-porous **Yttria-Stabilized Zirconia (YSZ)** (ZrO_2 doped with 8 mol% Y_2O_3). At high temperatures, oxygen vacancies in the cubic fluorite lattice facilitate rapid conduction of oxide ions (O^{2-}).
- **Electrode Materials:**
  - **Anode (Fuel Electrode):** Porous Nickel-YSZ cermet ($Ni-YSZ$). Nickel catalyzes hydrocarbon reforming and provides electronic conductivity.
  - **Cathode (Air Electrode):** Porous Lanthanum Strontium Manganite (LSM, $La_{1-x}Sr_xMnO_3$).
- **Electrode Reactions:**
  - **Cathode:** $O_2(g) + 4e^- \rightarrow 2O^{2-}$(migrates across YSZ ceramic)
  - **Anode (H_2 Fuel):** $2H_2(g) + 2O^{2-} \rightarrow 2H_2O(g) + 4e^-$
  - **Anode (CO Fuel):** $CO(g) + O^{2-} \rightarrow CO_2(g) + 2e^-$
  - **Overall:** $2H_2 + O_2 \rightarrow 2H_2O$ and $2CO + O_2 \rightarrow 2CO_2$
- **Advantages: Fuel flexibility** (can directly utilize natural gas, biogas, and syngas without external reformers); non-precious catalysts (no expensive Pt needed); combined heat and power (CHP) cogeneration efficiencies exceed $80-85\%$.
- **Disadvantages:** High operating temperature requires expensive ceramic interconnects; slow startup time; thermal cycling brittleness (unsuitable for passenger cars; ideal for stationary power plants).

7. Solar Energy Conversion & Dye-Sensitized Solar Cells (DSSC)

7.1 Photovoltaic (PV) vs. Photoelectrochemical (PEC) vs. DSSC

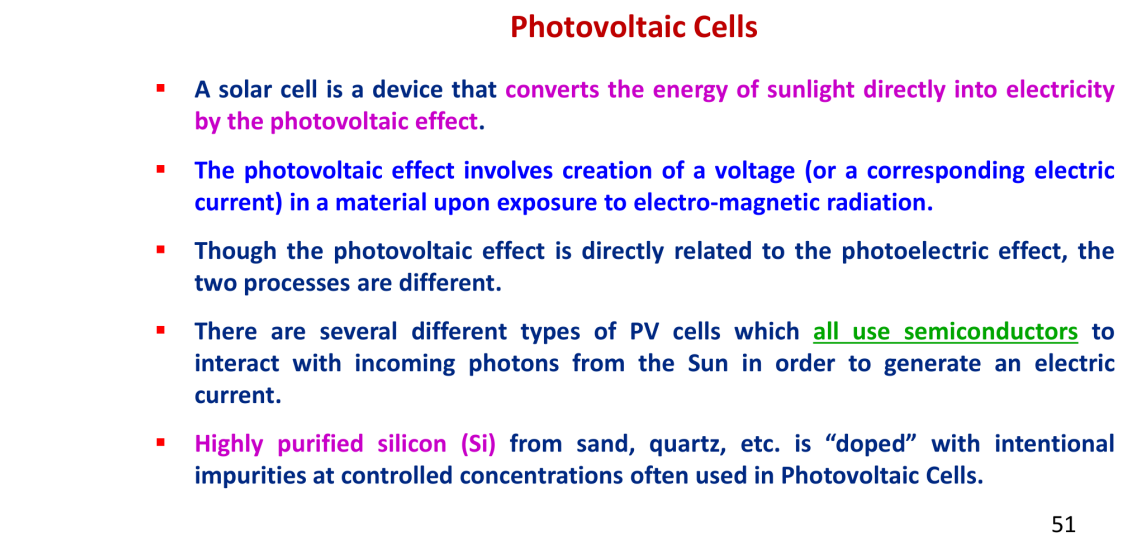


Figure 2.10: Photovoltaic conversion in silicon p-n junction solar cell.

Parameter	Silicon Photovoltaic (PV)	Photoelectrochemical (PEC)	Dye-Sensitized Solar Cell (DSSC)
Working Principle	p-n junction internal electric field separates photogenerated electron-hole pairs.	Semiconductor-electrolyte junction band bending drives redox or water splitting.	Molecular dye absorbs light; ultrafast electron injection into mesoporous TiO_2.
Light Absorption	Silicon semiconductor crystal absorbs photon across bandgap E_g.	Semiconductor electrode (TiO_2, Fe_2O_3) absorbs light.	Monolayer of chemisorbed organometallic or natural dye absorbs photons.
Electrolyte	Solid state throughout (no liquid electrolyte).	Liquid aqueous or non-aqueous electrolyte.	Liquid or gel redox electrolyte ($\frac{I^-}{I_3^-}$ redox couple).
Cost & Fabrication	High energy consumption; high-temperature cleanroom processing.	Moderate cost; electrode corrosion/photocorrosion challenges.	Low cost; simple screen-printing and ambient temperature roll-to-roll fabrication.
Efficiency	Commercial: $18\% - 24\%$; Lab: $> 26\%$.	Solar-to-chemical: $10\% - 15\%$.	Commercial: $10\% - 13\%$; Lab: 14.5%.

7.2 Architecture of the Grätzel Dye-Sensitized Solar Cell (DSSC)

Invented in 1991 by Brian O'Regan and Michael Grätzel, the DSSC mimics natural plant photosynthesis using a 5-layer sandwich structure:

1. **Transparent Conductive Substrate (TCO):** Fluorine-doped Tin Oxide (FTO) glass ($SnO_2 : F$) providing optical transparency ($> 85\%$) and high electrical conductivity.
2. **Mesoporous Semiconductor Film:** Nanocrystalline Titanium Dioxide (TiO_2, anatase phase, nanoparticle diameter ≈ 20 nm, film thickness $\approx 10\mu$ m) sintered at 450° C. It provides an enormous internal surface area (roughness factor > 1000) for dye adsorption.
3. **Photosensitizer Dye:** Chemisorbed as a self-assembled monolayer onto TiO_2 nanoparticles via anchoring carboxylic acid ($-COOH$) groups.
   - Organometallic complexes: Ruthenium polypyridyl dyes (N719 dye, N3 dye, Black dye).
   - Natural organic dyes: Anthocyanins extracted from blackberries, red cabbage, and pomegranate.

4. **Redox Electrolyte:** Iodide/Triiodide ($\frac{I^-}{I_3^-}$) redox shuttle dissolved in an organic nitrile solvent (acetonitrile).
5. **Counter Electrode (Cathode):** FTO glass coated with an ultrathin catalytic layer of Platinum (Pt) or carbon.

7.3 Four-Step Working Mechanism of DSSC

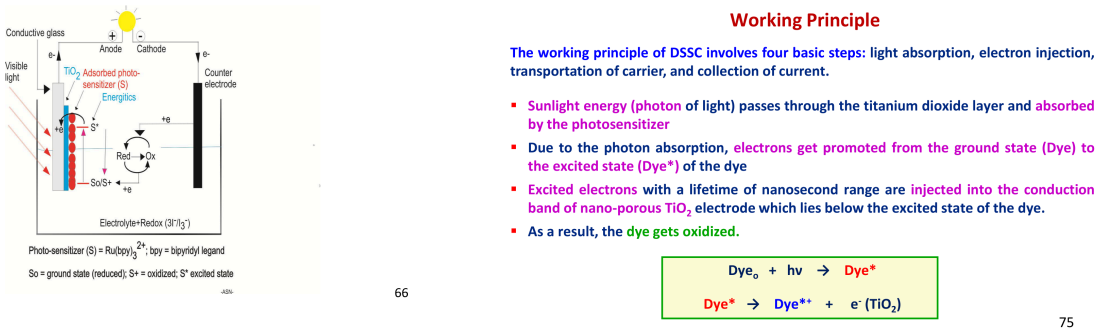


Figure 2.11: (Left) Dye-Sensitized Solar Cell 5-layer sandwich structure; (Right) 4-step photoelectric conversion mechanism and redox regeneration cycle.

- **Energy Level Hierarchy Requirement:**
  - Dye LUMO must be higher (more negative) than the Conduction Band edge of TiO_2 (-0.5 V vs NHE) to enable spontaneous electron injection.
  - Dye HOMO must be lower (more positive) than the redox potential of the $\frac{I^-}{I_3^-}$ couple ($+0.35$ V vs NHE) to ensure rapid regeneration of oxidized dye (D^+).
  - Open-circuit photovoltage (V_{oc}): Determined by the difference between the Fermi level of TiO_2 and the redox potential of the electrolyte ($V_{oc} \approx 0.7 - 0.9$ V).